AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM254
Managing IT: the why, the what and the how

Final Examination
Fall Semester 2024/2025

Date: 04 /Jan./2025

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Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1: MCQ [20 Marks]

Each question carries a weight of 2 marks

Question 1: [20 marks] Read the following questions and choose the best answer.

1- Which of the following is NOT part of the Iron Triangle in project management?

A. Cost
B. Quality
C. Time
D. Scope

2- Which category do users, sponsors, and other organization members belong to?

A. Team stakeholders
B. Organizational stakeholders
C. External stakeholders
D. Contracted stakeholders

3- _____ to software or systems development is used where the requirements are not fully known but there is a need to develop a prototype version of the proposed system.

A. Iterative approach
B. Waterfall approach
C. Agile approach
D. Incremental approach

4- A high-level description of a database that is detailed enough to be used by technical stakeholders and easily understood by non-technical stakeholders:

A. Entity Relationship Diagram
B. Database Schema
C. Data Dictionary
D. Conceptual Data Model

5- Which of the below states that everything that needs to be known is inside the database?

A. Closed World Assumption
B. Open World Assumption
C. Open Database
D. Closed Database

6- The operator that takes two sets and returns the set of values that are present in the first set but not in the second set is known as:

A. Difference Operator
B. Union Operator
C. Intersection Operator
D. Symmetric Operator

7- What does the Product Breakdown Structure (PBS) represent?

A. The detailed specifications of individual products
B. The hierarchical relationship between products and their subcomponents
C. The financial breakdown of product costs
D. The scheduling of product development phases

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8- When each non-key attribute is directly (non-transitively) dependent on the full key, this refers to:

A. First Normal Form
B. Second Normal Form
C. Third Normal Form
D. Fourth Normal Form

9- The work schedule for a project is derived from which two key components?

A. The project charter and stakeholder analysis
B. The activity bar chart and resource table
C. The project budget and risk assessment
D. The communication plan and quality management plan

10-Access control protects data from:

A. Authorized users
B. Unauthorized users
C. Both authorized and unauthorized users
D. Neither authorized nor unauthorized users

PART 2: Short questions [25 Marks]

Question 2: [6 marks]

A project has many characteristics – some relating to the application itself, others to the project structure and management– and several of them are interrelated. List six of the characteristics questions that will form the basis of the project approach.

Question 3: [4.5 marks]

Explain the difference between full data backup, incremental data backup, and differential data backup.

Question 4: [4.5 marks]

There are three types of anomalies that can occur in a database table. List and explain each type.

Question 5: [10 marks]

For each of the following relations, determine whether it is in Second Normal Form (2NF) and justify your answer. If a relation is not in 2NF, transform it into 2NF.

Note:" The primary keys are indicated in bold and underlined."

a- OrderDetail (**OrderID**, **ProductID**, ProductName, Quantity)
OrderDetail (001, P001, Laptop, 2)
OrderDetail (001, P002, Mouse, 5)
OrderDetail (002, P001, Laptop, 1)
OrderDetail (003, P003, Keyboard, 3)

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b- EmployeeProjects (**EmployeeID**, **ProjectID**, ProjectName, HoursWorked)
EmployeeProjects (1, (201, 202), (Website Design, Database Setup),35)
EmployeeProjects (2, (201, 203), (Website Design, Security Review),40)

Part 3: Problem Solving Questions [55 Marks]

Question 6: [15 marks]

The **Employees** table below stores basic information about employees in an organization. The table includes the following columns: EmployeeID (a unique identifier for each employee), Name (the employee's name), Department (the department they belong to), Salary (their current salary), and JoiningDate (the date they joined the company).

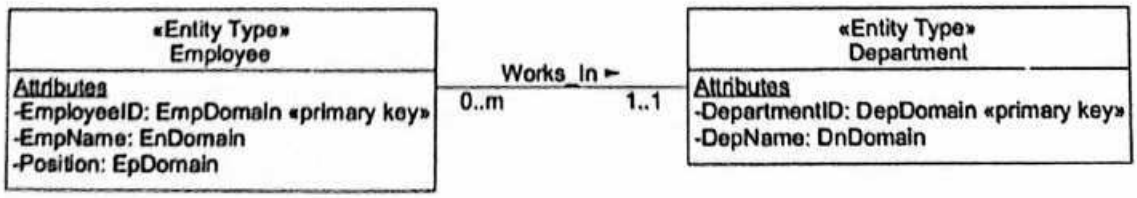
Table: Employees

EmployeeID	Name	Department	Salary	JoiningDate
1	John Smith	HR	5000	6/15/2020
2	Alice Brown	IT	6000	1/20/2021
3	Bob White	HR	5500	5/10/2019
4	Carol Green	IT	6500	3/5/2022

a) Write a query that selects the names and salaries of employees from the IT department, where the salary is greater than 6000, and sorts by salary in descending order. (3 marks)
b) Write a query that updates the salary of employee EmployeeID: 3 to 5800. (2 marks)
c) Write a query that selects the average salary of employees in each department, but only includes departments where the average salary is greater than 5500. Use GROUP BY and HAVING. (4 marks)
d) Write a query that creates a view showing employee names and their departments. (3 marks)
e) Write a query that creates a new user called viewer_user and grants this user access to the view created in the previous question. (3 marks)

Question 7: [20 marks]

The below E-R diagram shows the relationship **Works_In** between the Employee and Department entity types:



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a) Write the text form of each entity type including its attributes and domains. (4 Marks)
b) What is the optionality and multiplicity of **Employee** in the **Works_In** relationship? (2 Marks)
c) Draw the relation-for-relationship representation from the original E-R diagram. (8 Marks)
d) Draw the posted-key representation suitable from the original E-R diagram. (6 marks)

Question 8: [20 marks]

A team is working on developing an e-commerce website. The needed activities are shown below, noting that they want to work on Backend Analysis and Interface Design simultaneously

Activity	Duration (days)
Project Requirements	3 days
Backed Development	2 days
Testing interface and Backend	3 days
Interface design	5 days
Backend Analysis	4 days

a) Draw an Activity-on-Node (AoN) diagram to visualize the development process of the mobile application. (5 marks)
b) Add the duration, earliest start, and earliest values to each node. (7.5 marks)
c) Add the Latest Start, earliest start, and Float to each node. (7.5 marks)

End of questions

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